



Course Code and Number: CS-UH 3010
Transcript Course Title: Operating Systems
Semester: Spring 2025

Meeting time and location	Day/Time	Location
Lecture	M/5.00 PM – 6.15PM	C2 E050
Lecture	W/5.00 PM – 6.15PM	C2 E050
Recitation	F/10.50AM – 12.05PM	Campus Center E047
Instructional time requirements are met by the scheduled meeting times.		

Faculty Details	
Name	Eliseo Ferrante
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Telephone	8-4934
Workspace	A2-195
Office Hours	Tuesday afternoon

Credits: 3

Prerequisites: CS-UH1052 and CS-UH2010

Corequisites: N/A

This course counts toward the following NYUAD degree requirements:

Majors > Computer Science

Minors > Computer Science

Majors > Computer Engineering

Course Description

An Operating System (OS) helps users develop, run, and store their programs and is the intermediary between the human using a machine and the computer hardware. The main objective of the course is to present fundamental concepts and outline principles upon which modern Operating Systems are designed and implemented. Internal components that manage both hardware and logical resources and their cooperation are outlined. Key

mechanisms and requisite policies that govern OS modules along with pertinent algorithmic aspects are introduced. Emphasis is placed on how programmatic and command interfaces assist users to both successfully and effectively run their applications without depleting OS vital resources.

The course covers classic concepts including synthesis of computing systems, processes, user/kernel-mode of execution, batch processing, time sharing, multi-programming, and DMA-enabled-IO operations. We present concurrent processes, the critical section problem, mechanisms that enable the correct execution of concurrent tasks, inter-process communication primitives, CPU scheduling, deadlocks, and threads. We outline management techniques for main memory, virtual memory and corresponding policies, storage structures, file system interfaces and their implementation as well as basic integrity mechanisms for file systems.

The study of the subject is balanced with appropriate programming projects in the Linux Operating Systems and a number of homework assignments throughout the semester.

Course Learning Outcomes (CLOs) and Links to Program Learning Outcomes (PLOs)

Upon successful completion of this course the student will be able to:	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6
CLO1: Produce solutions for modern system development.	Med	Med				
CLO2: Apply computer systems techniques and respective APIs to create OS modules.	High	High	High			
CLO3: Represent and model concepts in systems building terms.	High		Med		Med	
CLO4: Apply tools and methods taught in theoretical terms in computer science courses.			High	Med		
CLO5: Ascertain and experimentally evaluate trade-offs when synthesis of OS modules is realized.	High				Med	Med

Teaching and Learning Methodologies

- 1. Direct Instruction in the form of Lectures, Questions, in-class Discussions.** During each class, there will be numerous questions posed by the instructor to help students engage in the topics and to promote discussion.
- 2. Directed Self-Learning by Reading.** Lecture slides are made available to the students. The students are also advised read and study material from the textbooks.
- 3. Writing.** Students will represent concepts in the context of computing systems, write solutions to problems and derive design choices for salient aspects of contemporary Operating Systems.
- 4. Pragmatic Programming Development and Pertinent Problem Solving.** Programming assignments will help students better understand the principles covered in class with

programming projects and offer solutions whose design tradeoffs are well understood, corroborated in writing and ultimately presented in oral presentations.

INSTRUCTIONAL ACTIVITIES	Format	Minutes	Frequency	Total Time
Lectures	in-person	75	28	2100
Labs (counting 0.5 for instructional time)	in-person	75	10	375
Final Exam during Exam Period	in-person	120	1	120
TOTAL				2595

Graded Activities

	Grade Percentage	Submission Date	Linked to Course Learning Outcome(s)
Homework Assignments	05%	Ongoing	1, 2, 3
Programming Projects	35%	Week 6, Week 9, Week 12, Week 15	2, 3, 4
Midterm Exam	20%	Week 8	1, 2, 5
Final Exam	40%	Week 15	1, 2, 5

Description of Graded Activities

The homework assignment follows from lab sessions. They are used to have an immediate assessment of the degree of learning of the tools and techniques learned in the lab.

Programming projects will have four deliveries during the course: students will apply theoretical knowledge and tools learned in the lab to design components of an operating system. For these projects, students will work in groups of two and present their finding.

Midterm exam, held before the seasonal break, is held during class, and assesses the students on the theoretical concepts acquired up to that point.

Final exam is cumulative on the whole syllabus and is held during the final's week.

Grade Distribution

Students need to obtain a grade of C or better to count the course towards their intended degree for required courses or economics electives. Course percentages will be translated into letter grades based on these intervals:

A	A-	B+	B	B-	C+	C	C-	D+	D	F
[100-95]	[94-90]	[89-87]	[86-83]	[82-80]	[79-77]	[76-73]	[72-70]	[69-67]	[67-63]	[62-0]

Course Materials

Required Texts to be purchased by the bookstore

- A. Silberschatz, P.B. Galvin, and G. Gagne, Operating System Concepts, 10th Edition, John Wiley & Sons Inc., New York, NY, December 2019.

Optional Readings

- R. Arpaci-Dusseau and A. Arpaci-Dusseau, Operating Systems: Three Easy Pieces, Arpaci-Dusseau Books, LLC, 2023 (Version 1.10).
<https://pages.cs.wisc.edu/~remzi/OSTEP/>
- A. S. Tannenbaum, Modern Operating Systems, Fifth Edition. Pearson, 2022.
- W. Stallings, Operating Systems: Internals and Design Principles, Ninth Edition. Pearson, 2018
- M. Fragouli, D. Leventis, A. Petropoulos and A. Delis, Working with the Unix OS, 2011
<http://www.alexdelis.eu/k22/WorkingWithUnix.pdf>

Academic Policies

Attendance

Class attendance for this core course is expected. Students should make every effort to attend classes and remain in close touch with the material and the discussions taking place throughout the semester. If one has to be absent for any reason, the instructor has to be notified in *writing beforehand*.

Class Participation

Class participation and student questions are strongly encouraged as they invariably help better understand the material.

Artificial Intelligence

Use of ChatGPT and related tools is allowed in this class, but only in the ways recommended in the lecture notes and during discussions in class and in the lab. When you use any of these tools, you must include a note describing how you used them with the assignment. As a general rule, you are not allowed to use AI-generated content that you do not understand, and you are strongly encouraged to “criticize” the generated content and to include it in a way that you fully control.

Integrity

At NYU Abu Dhabi, a commitment to excellence, fairness, honesty, and respect within and outside the classroom is essential to maintaining the integrity of our community. By accepting membership in this community, students, faculty, and staff take responsibility for demonstrating these values in their own conduct and for recognizing and supporting these values in others. In turn, these values create a campus climate that encourages the free exchange of ideas, promotes scholarly excellence through active and creative thought, and allows community members to achieve and be recognized for achieving their highest potential.

Students should be aware that engaging in behaviors that violate the standards of academic

integrity will be subject to review and may face the imposition of penalties in accordance with the procedures set out in the NYUAD policy:

<https://students.nyuad.nyu.edu/academics/registration/academic-policies/academic-integrity/>

NYU Moses Center for Student Accessibility: New York University is committed to providing equal educational opportunity and participation for students with disabilities. The center works with NYU students to determine appropriate and reasonable accommodations that support equal access to a world-class education. Confidentiality is of the utmost importance. Disability-related information is never disclosed without student permission.

Find further information at:

<https://students.nyuad.nyu.edu/student-affairs/student-persistence/students-with-disabilities/>

Contact: mosescsa@nyu.edu (this emails the center at NYU directly)

nyuad.mosescenterliasion@nyu.edu (this emails the NYUAD team)

Mental Health Resources

As a University student, you may experience a range of issues that can interfere with your ability to perform academically or impact your daily functioning, such as: heightened stress; anxiety; difficulty concentrating; sleep disturbance; strained relationships; grief and loss; personal struggles. If you have any well-being or mental health concerns please visit the Counseling Center on the ground floor of the campus center from 9am-5pm Abu Dhabi time Sunday - Thursday, or schedule an appointment to meet with a counselor by calling: +971 02-628-8100, or emailing: nyuad.healthcenter@nyu.edu. If you require mental health support outside of these hours, call NYU's Wellness Exchange hotline at +971 02-628-5555, which is available 24 hours a day, 7 days a week. You can also utilize the Wellness Exchange mobile chat feature, details of which you can find on the student portal.

Course Schedule

Week	Date	Day	Topic	Reading	Online
1	1/22	W	Introduction to the course	Chapter 1	
2	1/27 1/29	M W	Classic Concepts in OSES Operating Systems Services	Chapter 1 Chapter 2	
3	2/03 2/05	M W	Operating systems Structures Processes and Operation on Processes	Chapter 2 Chapter 3	
4	2/10 2/12	M W	Inter Process Communication (IPC) shared memory IPC: Message passing, sockets	Chapter 3	
5	2/17 2/19	M W	Threads, threading models Threading libraries, implicit threading	Chapter 4	
6	2/24 2/26	M W	CPU Scheduling, criteria CPU Schedulers, evaluation of schedulers	Chapter 5	
7	3/03 3/05	M W	Midterm review Midterm	Chapters 1 to 5	
8	3/10 3/12	M W	TBD (Mild intro to next topic, guest lecture, etc.) No classes as per the Academic Calendar		
9	3/17 3/19	M W	Synchronization: Critical Section Mutex, semaphores, monitors	Chapter 6	
10	3/24 3/26	M W	Classic synchronization examples Dining philosophers	Chapter 7	
11	3/31 4/02	M W	Eid Al Fitr – Spring Break Eid Al Fitr – Spring Break		
12	4/07 4/09	M W	Deadlocks: characterization and prevention Deadlock avoidance and detection	Chapter 8	
13	4/14 4/16	M W	Main memory: addresses, contiguous allocation Fragmentation, paging, page table structure	Chapter 9	
14	4/21 4/23	M W	Virtual memory: demand paging Page replacement, frame allocation, thrashing	Chapter 10	
15	4/28 4/30	M W	I/O: types and interfaces I/O: subsystems and performance	Chapter 12	
16	5/05 5/07	M W	File system implementation Final review	Chapter 14 All	
17	5/12 5/16		Final Exam	All	

Appendix 1

CS Major Program Learning Outcomes (PLOs)

PLO 1: Analyze a problem, and identify, define, and verify the appropriate computational tools required to solve it.

PLO 2: Apply up-to-date computational tools necessary in a variety of computing practices.

PLO 3: Implement algorithms as programs using modern computer languages.

PLO 4: Apply their mathematical knowledge to solve computational problems.

PLO 5: Communicate computer science knowledge both orally and in writing.

PLO 6: Collaborate in teams.